

Continuous Assessment II

Paul Haskins

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Question 1

0.0.1 Part 1

We have the expression for entropy in the grand canonical ensemble

$$S = - \left(\frac{\partial J}{\partial T} \right)_{V, \mu} \quad (1)$$

with

$$J = -kT \ln \Xi \quad (2)$$

We know from lectures that we have an expression for the logarithm of the grand canonical partition function

$$\ln \Xi = -\ln(1-z) + (\deg) V \left(\frac{2\pi m}{\hbar^2 \beta} \right)^{\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(z) \quad (3)$$

We can define a new constant

$$\alpha = (\deg) V \left(\frac{2\pi m}{\hbar^2} \right)^{\frac{3}{2}} \quad (4)$$

$$\ln \Xi = -\ln(1-z) + \alpha \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(z) \quad (5)$$

$$\ln \Xi = -\ln(1-e^{\beta\mu}) + \alpha \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(e^{\beta\mu}) \quad (6)$$

Then we can write the entropy as

$$S = - \left[\frac{\partial \beta}{\partial T} \frac{\partial}{\partial \beta} \ln \Xi \right] = -\frac{J}{T} + kT \frac{\partial \ln \Xi}{\partial \beta} \quad (7)$$

We can use the property of the polylogarithms that

$$z \frac{d}{dz} \mathcal{L}i_r(z) = \mathcal{L}i_{r-1}(z) \quad (8)$$

$$S = - \left[\frac{\partial \beta}{\partial T} \frac{\partial}{\partial \beta} \ln \Xi \right] = -\frac{J}{T} + kT \left[\frac{e^{\beta\mu}}{1-e^{\beta\mu}} - \frac{3}{2} \alpha \beta^{-\frac{5}{2}} \mathcal{L}i_{\frac{5}{2}}(e^{\beta\mu}) + \alpha \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(e^{\beta\mu}) \mu \right] \quad (9)$$

$$S = k \left[-\ln(1-z) + \frac{5}{2} \alpha \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(z) \right] + k \left[\frac{-\beta \mu e^{\beta\mu}}{1-z} + \frac{3}{2} \alpha \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(z) - \alpha \mu \beta^{-\frac{1}{2}} \mathcal{L}i_{\frac{3}{2}}(z) \right] \quad (10)$$

Notice $\ln(z) = \mu\beta$.

Thus we have

$$S = k \left[-\ln(1-z) + \frac{5}{2} \alpha \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(z) \right] + k \left[\frac{-z \ln z}{1-z} + \frac{3}{2} \alpha \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(z) - \alpha \mu \beta^{-\frac{1}{2}} \mathcal{L}i_{\frac{3}{2}}(z) \right] \quad (11)$$

We can write in terms of our initial constants as

$$S = k \left[-\ln(1-z) - \frac{z \ln(z)}{1-z} - V(\deg) \left(\frac{2\pi m}{\hbar^2} \right) \left(\ln(z) \mathcal{L}i_{\frac{3}{2}}(z) - \frac{5}{2} \mathcal{L}i_{\frac{5}{2}}(z) \right) \right] \quad (12)$$

Now we examine the limiting cases where we have $T \ll T_c$ then $z \approx 1$.

$$-V(\deg) \left(\frac{2\pi m}{\hbar^2} \right) \left(\ln(z) \mathcal{L}i_{\frac{3}{2}}(z) - \frac{5}{2} \mathcal{L}i_{\frac{5}{2}}(z) \right) \approx -V(\deg) \left(\frac{2\pi m}{\hbar^2} \right) \left(-\frac{5}{2} \mathcal{L}i_{\frac{5}{2}}(1) \right) \quad (13)$$

as $\ln 1 = 0$ $\mathcal{L}i_{\frac{5}{2}}(1) = \text{const.}$ And we have

$$\lim_{z \rightarrow 1} \frac{z \ln(z)}{1-z} = -1 \quad (14)$$

$$\lim_{z \rightarrow 1} \ln(1-z) = -\infty \quad (15)$$

$S \approx -k \ln(1-z)$ for $T \ll T_c$.

Now if we take $T \gg T_c$ we $z \ll 1$.

We can use the summation definition of the poly log to take leading order i.e for $z \ll 1$.

$$\mathcal{L}i_r(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^r} \approx z \quad (16)$$

so for large z

$$\frac{5}{2} \mathcal{L}i_{\frac{5}{2}}(z) \ll -\ln(z) \mathcal{L}i_{\frac{3}{2}}(z) \quad (17)$$

$$S \approx -kz \ln(z) \left[\frac{1}{1-z} + V(\text{deg}) \left(\frac{2\pi m}{\hbar^2 \beta} \right)^{3/2} \right] \quad (18)$$

with $\frac{1}{1-z} \approx 1$ and $\beta^{-\frac{3}{2}} \gg 1$

$$S \approx -kz \ln(z) V(\text{deg}) \left(\frac{2\pi m}{\hbar^2 \beta} \right)^{3/2} \quad (19)$$

0.0.2 (Part 2)

Previously we had

$$S = k \left[-\ln(1-z) - \frac{z \ln(z)}{1-z} - V(\text{deg}) \left(\frac{2\pi m}{\hbar^2} \right) \left(\ln(z) \mathcal{L}i_{\frac{3}{2}}(z) - \frac{5}{2} \mathcal{L}i_{\frac{5}{2}}(z) \right) \right] \quad (20)$$

With the following relations for the average number of particles and the average energy

$$\langle N \rangle = z \frac{\partial \Xi}{\partial z}, \langle E \rangle = \mu \langle N \rangle - \frac{\partial \ln \Xi}{\partial \beta} \quad (21)$$

where we already calculated

$$\frac{\partial \ln \Xi}{\partial \beta} = \left[\frac{e^{\beta \mu}}{1 - e^{\beta \mu}} - \frac{3}{2} \alpha \beta^{-\frac{5}{2}} \mathcal{L}i_{\frac{5}{2}}(e^{\beta \mu}) + \alpha \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(e^{\beta \mu}) \mu \right] \quad (22)$$

$$\langle N \rangle = \frac{z}{1-z} + a \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(z) = \frac{z}{1-z} + (\text{deg}) V \left(\frac{2\pi m}{\hbar^2} \right)^{\frac{3}{2}} \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(z) \quad (23)$$

and

$$\langle E \rangle = \frac{3}{2} (\text{deg}) V \left(\frac{2\pi m}{\hbar^2} \right)^{\frac{3}{2}} \beta^{-\frac{5}{2}} \mathcal{L}i_{\frac{5}{2}}(z) \quad (24)$$

Thus we can identify

$$S = k \left[-\ln(1-z) - \ln z (\langle N \rangle) + \frac{5\beta}{3} \langle E \rangle \right] \quad (25)$$

0.0.3 (Part 3)

$$N = \langle n_0 \rangle + N_{exc} \quad (26)$$

$$N = \frac{z}{1-z} + V(\text{deg}) \left(\frac{2\pi m}{\hbar^2 \beta} \right)^{\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(z) \quad (27)$$

We can identify

$$\langle n_0 \rangle = \frac{z}{1-z}, N_{exc} = V(\text{deg}) \left(\frac{2\pi m}{\hbar^2 \beta} \right)^{\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(z) \quad (28)$$

Now we can examine the behaviour in limiting cases In the regime where $T \ll T_c$ we have $\langle n_0 \rangle \propto N$

$$\langle n_0 \rangle = \frac{z}{1-z} \quad (29)$$

$$z = \frac{1}{1 + \frac{1}{\langle n_0 \rangle}} \propto \frac{1}{1 + \frac{1}{N}} \quad (30)$$

$$= \frac{1}{1 + \frac{1}{N}} \quad (31)$$

Using the geometric series we have

$$z = 1 - \frac{1}{N} + O\left(\frac{1}{N^2}\right) \quad (32)$$

Thus we get

$$S \approx -k \ln\left(\frac{1}{N}\right) = k \ln N \quad (33)$$

In the regime where $T \gg 1$ we have that $z = e^{\beta\mu} \ll 1$, $N_{ex} \approx N$ aswell as $\langle n_0 \rangle \ll 1$

$$N_{ex} = V(\text{deg}) \left(\frac{2\pi m}{h^2 \beta} \right)^{\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(z) \quad (34)$$

$$N \approx V(\text{deg}) \left(\frac{2\pi m}{h^2 \beta} \right)^{\frac{3}{2}} z \quad (35)$$

Thus we get our entropy

$$S \approx -kN \ln z \quad (36)$$

We know

$$\langle n_0 \rangle = N - N_{ex} = N - V(\text{deg}) \left(\frac{2\pi m}{h^2 \beta} \right)^{\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(z) \quad (37)$$

Which can be written as

$$= N - \left[\frac{\hbar^2}{2\pi m k} \left(\frac{N}{V(\text{deg}) \mathcal{L}i_{\frac{3}{2}}(1)} \right)^{\frac{2}{3}} \right]^{-\frac{3}{2}} NT^{\frac{3}{2}} \frac{\mathcal{L}i_{\frac{3}{2}}(z)}{\mathcal{L}i_{\frac{3}{2}}(1)} \quad (38)$$

We recognize the expression for the critical temperature and can use the definition of the poly logarithm to get

$$\approx \left[1 - \left(\frac{T}{T_c} \right)^{\frac{3}{2}} \frac{z}{\zeta\left(\frac{3}{2}\right)} \right] \ll 1 \quad (39)$$

$$z \approx \zeta\left(\frac{3}{2}\right) \left(\frac{T_c}{T} \right)^{\frac{3}{2}} \quad (40)$$

Thus our entropy is

$$S \approx -kN \ln \left[\zeta\left(\frac{3}{2}\right) \left(\frac{T_c}{T} \right)^{\frac{3}{2}} \right] \quad (41)$$

$$= kN \left[\frac{3}{2} \ln \left(\frac{T}{T_c} \right) - \ln \left(\zeta\left(\frac{3}{2}\right) \right) \right] \quad (42)$$

$$S \approx kN \left[\frac{3}{2} \ln \left(\frac{T}{T_c} \right) \right] \quad (43)$$

Question 2

0.0.4 (I)

We are given

$$P = \frac{2}{3} \langle E \rangle \frac{1}{V} = \frac{3}{2} \left(\frac{2\pi mk}{h^2} \right)^{\frac{3}{2}} T^{\frac{5}{2}} \mathcal{L}i_{\frac{5}{2}}(z) \quad (44)$$

$$\left(\frac{\partial P}{\partial T} \right)_P = 0 \quad (45)$$

$$0 = \frac{3}{2} \left(\frac{2\pi mk}{h^2} \right)^{\frac{3}{2}} \left[\frac{5T^{\frac{3}{2}}}{2} \mathcal{L}i_{\frac{5}{2}}(z) + \frac{T^{\frac{5}{2}}}{z} z \left(\frac{\partial \mathcal{L}i_{\frac{5}{2}}(z)}{\partial z} \right)_P \left(\frac{\partial z}{\partial T} \right)_P \right] \quad (46)$$

Rearranging gives the required result

$$\frac{1}{z} \left(\frac{\partial z}{\partial T} \right)_P = -\frac{5}{2T} \frac{\mathcal{L}i_{\frac{5}{2}}(z)}{\mathcal{L}i_{\frac{3}{2}}(z)} \quad (47)$$

For the second equation we can use the fact that

$$N = \langle N \rangle \quad (48)$$

$$N \approx V(\text{deg}) \left(\frac{2\pi mk}{h^2} \right)^{\frac{3}{2}} T^{\frac{3}{2}} \mathcal{L}i_{\frac{3}{2}}(z) \quad (49)$$

$$\left(\frac{\partial N}{\partial T} \right)_N = \left(\frac{\partial N}{\partial T} \right)_V \quad (50)$$

$$= V(\text{deg}) \left(\frac{2\pi mk}{h^2} \right)^{\frac{3}{2}} \left[\frac{3}{2} T^{\frac{1}{2}} \mathcal{L}i_{\frac{3}{2}}(z) + \frac{T^{\frac{3}{2}}}{z} z \left(\frac{\partial \mathcal{L}i_{\frac{3}{2}}(z)}{\partial z} \right)_V \left(\frac{\partial z}{\partial T} \right)_V \right] \quad (51)$$

Rearranging we retrieve the required result

$$\frac{1}{z} \left(\frac{\partial z}{\partial T} \right)_V = -\frac{3}{2T} \frac{\mathcal{L}i_{\frac{3}{2}}(z)}{\mathcal{L}i_{\frac{1}{2}}(z)} \quad (52)$$

0.0.5 (II)

By definition

$$\gamma = \frac{C_P}{C_V} \quad (53)$$

with the following two definitions

$$C_P = T \left(\frac{\partial S}{\partial T} \right)_{N,P} = T \left(\frac{\partial S}{\partial z} \right)_N \left(\frac{\partial z}{\partial T} \right)_P \quad (54)$$

$$C_V = T \left(\frac{\partial S}{\partial z} \right)_{N,V} = T \left(\frac{\partial S}{\partial z} \right)_N \left(\frac{\partial z}{\partial T} \right)_V \quad (55)$$

Thus we have

$$\gamma = \frac{T \left(\frac{\partial S}{\partial z} \right)_N \left(\frac{\partial z}{\partial T} \right)_P}{T \left(\frac{\partial S}{\partial z} \right)_N \left(\frac{\partial z}{\partial T} \right)_V} \quad (56)$$

$$\gamma = \frac{\frac{1}{z} \left(\frac{\partial z}{\partial T} \right)_P}{\frac{1}{z} \left(\frac{\partial z}{\partial T} \right)_V} = \frac{5}{3} \frac{\mathcal{L}i_{\frac{5}{2}}(z) \mathcal{L}i_{\frac{1}{2}}(z)}{\left(\mathcal{L}i_{\frac{3}{2}}(z) \right)^2} \quad (57)$$

Question 3

I am assuming it is sufficient to take the following expression for $\ln \Xi$ as a given as it was provided in the tutorial. There is a very rough derivation provided at the end of this PDF but i assume it isn't required.

$$\ln \Xi = -\ln(1-z) - (\deg) \frac{L^d}{h^d} S_{d-1} \frac{1}{2} (2m)^{\frac{d}{2}} \int_0^\infty d\epsilon \left(\epsilon^{\frac{d}{2}-1} \right) \ln(1 - z e^{-\beta\epsilon}) \quad (58)$$

$$\langle n \rangle = z \frac{\partial \ln \Xi}{\partial z} = \frac{z}{1-z} - (\deg) \frac{L^d}{h^d} S_{d-1} \left(\frac{1}{2} \right) (2m)^{\frac{d}{2}} \int_0^\infty d\epsilon \epsilon^{\frac{d}{2}-1} \frac{1}{\frac{e^{\beta\epsilon}}{z} - 1} \quad (59)$$

We notice

$$\int_0^\infty d\epsilon \epsilon^{\frac{d}{2}-1} \frac{1}{\frac{e^{\beta\epsilon}}{z} - 1} = \mathcal{L}i_{\frac{d}{2}}(z) \Gamma\left(\frac{d}{2}\right) \quad (60)$$

So we have for d dimensions

$$\langle n \rangle = \langle n_0 \rangle + (\deg) \frac{L^d}{h^d} \mathcal{L}i_{\frac{d}{2}}(z) \quad (61)$$

if we let $d = 1$

$$\langle n \rangle = \langle n_0 \rangle + (\deg) \frac{L}{h} \mathcal{L}i_{\frac{1}{2}}(z) \quad (62)$$

If we let $d = 2$ for the 2d case

$$\langle n \rangle = \langle n_0 \rangle + (\deg) \frac{L^2}{h^2} \mathcal{L}i_1(z) \quad (63)$$

In order for Bose-Einstein Condensation to occur we need an upper bound on the number of particles in excited states when our fugacity is $z \approx 1$ but we have that for the d dimensional case that the excited states are proportional to $\mathcal{L}i_{\frac{d}{2}}(1)$ which has a zeta function form of $\zeta\left(\frac{d}{2}\right)$. This zeta function doesn't converge for $d = 1$ or $d = 2$. However for $d = 3$ we get $\zeta\left(\frac{3}{2}\right)$ which does converge. This is why it occurs in 3D.

Question 4

0.0.6 (I)

$$\Xi = \prod_i \left(\sum_{n_i} e^{-\beta n_i (\epsilon_i - \mu)} \right) \quad (64)$$

We know the finite geometric series is given by

$$\sum_{i=1}^N r^i = \frac{r^{N+1} - 1}{r - 1} \quad (65)$$

let $x = e^{-\beta(\epsilon_i - \mu)}$

$$\langle n \rangle = z \frac{\partial \ln \Xi}{\partial z} = z \frac{\partial}{\partial z} \ln \left(\frac{x^{l+1} - 1}{x - 1} \right) \quad (66)$$

$$\langle n \rangle = z \frac{\partial}{\partial z} \ln(x^{l+1} - 1) - \ln(x - 1) \quad (67)$$

$$= z \left[\frac{(l+1) e^{-\beta\epsilon_i(l+1)} z^l}{z^l e^{-\beta\epsilon_i} - 1} - \frac{e^{-\beta\epsilon_i}}{z e^{-\beta\epsilon_i} - 1} \right] = \left[\frac{(l+1) e^{-\beta\epsilon_i(l+1)}}{e^{-\beta\epsilon_i} - \frac{1}{z^{l+1}}} - \frac{e^{-\beta\epsilon_i}}{e^{-\beta\epsilon_i} - \frac{1}{z}} \right] \quad (68)$$

$$\langle n \rangle = \frac{l+1}{\frac{e^{\beta\epsilon_i(l+1)}}{z^{l+1}} - 1} + \frac{1}{\frac{e^{\beta\epsilon_i}}{z} - 1} \quad (69)$$

0.0.7 (II)

We are given

$$\Xi = \prod_i \left(\sum_{n_i} e^{-\beta n_i (\epsilon_i - \mu)} \right) \quad (70)$$

$$\ln \Xi = \sum \ln(\zeta_i) \approx \int_0^\infty d\epsilon \omega_c(1, \epsilon) \ln \zeta_i \quad (71)$$

$$\zeta = \sum_{n_i=0}^l e^{-\beta n_i (\epsilon_i - \mu)} \quad (72)$$

$$\ln \Xi = \int_0^\infty d\epsilon a \sqrt{\epsilon} \ln \left(\sum_{n_i=0}^l e^{-\beta n_i (\epsilon_i - \mu)} \right) \quad (73)$$

We can use the finite series expansion to get

$$= \int_0^\infty d\epsilon a \sqrt{\epsilon} \left[\ln \left(\frac{x^{l+1} - 1}{x - 1} \right) \right] \quad (74)$$

Using log rules

$$= \int_0^\infty d\epsilon a \sqrt{\epsilon} \left[\ln(x^{l+1} - 1) - \ln(x - 1) \right] \quad (75)$$

We can use the taylor expansion of $\ln(1 - x)$

$$= \int_0^\infty d\epsilon a \sqrt{\epsilon} \left[- \sum_{n=0}^\infty \frac{x^{(l+1)n}}{n} + \frac{x^n}{n} \right] \quad (76)$$

X

$$= \sum_{n=0}^\infty \frac{a}{n} \int_0^\infty d\epsilon \sqrt{\epsilon} e^{-\beta(\epsilon - \mu)(l+1)n} + e^{-\beta(\epsilon - \mu)n} d\epsilon \sqrt{\epsilon} \quad (77)$$

$$\sum_{n=0}^\infty \frac{a z^{n(l+1)}}{n} \left[- \int_0^\infty d\epsilon \sqrt{\epsilon} e^{-\beta \epsilon (l+1)n} + z^n \int_0^\infty d\epsilon \sqrt{\epsilon} e^{-\beta \epsilon n} \right] \quad (78)$$

We see that we have an integral of the form

$$I(\tau) = \int_0^\infty d\epsilon \epsilon^{\frac{1}{2}} e^{-\beta \epsilon \tau} \quad (79)$$

If we make the sub $y = \beta \epsilon \tau$

$$I = \int_0^\infty \frac{dy}{\beta \tau^{\frac{3}{2}}} y^{\frac{1}{2}} e^{-y} \quad (80)$$

Which is an integral of the form of a gamma function

$$= \frac{1}{\beta \tau^{\frac{3}{2}}} \Gamma\left(\frac{3}{2}\right) \quad (81)$$

If we use this with our two definitions of τ above we obtain

$$\ln \Xi = \sum_{n=0}^\infty \frac{a}{n} \left[-z^{(l+1)n} \left[\frac{1}{\beta [(l+1)n]^{\frac{3}{2}}} \Gamma\left(\frac{3}{2}\right) + \left[\frac{z^n}{\beta n^{\frac{3}{2}}} \right] \right] \right] \quad (82)$$

We are only interested in wether it is proportional

$$\ln \Xi \propto \sum_{n=0}^\infty \left[-z^{(l+1)n} n^{-\frac{5}{2}} + z^n n^{-\frac{5}{2}} \right] \quad (83)$$

$$\propto -(l+1)^{-\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(z^{l+1}) + \mathcal{L}i_{\frac{5}{2}}(z) \quad (84)$$

We can use the fact that $\langle N \rangle = z \frac{\partial \ln \Xi}{\partial z}$.

$$\langle N \rangle \propto z \left[-(l+1)^{-\frac{1}{2}} \frac{\mathcal{L}i_{\frac{3}{2}}(z^{l+1}) z^l}{z^{l+1}} \right] + \mathcal{L}i_{\frac{3}{2}}(z) \quad (85)$$

So we get the required result

$$\langle N \rangle \propto -(l+1)^{-\frac{1}{2}} \mathcal{L}i_{\frac{3}{2}}(z^{l+1}) + \mathcal{L}i_{\frac{3}{2}}(z) \quad (86)$$

The image shows a chalkboard with handwritten mathematical derivations. The first line is $w(E) = \delta(E) + (deg) \int_{-\infty}^{\infty} d\epsilon \delta(E - \frac{\hbar^2 \epsilon^2}{2mL^2})$. The second line is $= \frac{L^d S_{d-1}(deg)}{h^d} \int_0^{\infty} dp p^{d-1} \delta(E - \frac{\hbar^2 p^2}{2m})$. The third line is $= \frac{L^d S_{d-1}(deg)}{h^d} \int_0^{\infty} dt \frac{\sqrt{2m}}{2} t^{-\frac{1}{2}} (\sqrt{2m} t)^{d-1} \delta(E - t)$. The fourth line is $= \frac{L^d S_{d-1}(deg)}{h^d} \frac{(2m)^{\frac{d}{2}}}{2} \int_0^{\infty} dt t^{\frac{d}{2}-1} \delta(E - t)$. The final result is $E^{\frac{d}{2}-1}$.

Figure 1: Here we have some rough work of the preliminaries of the solution to **question 3** . This work was done in lectures explicitly for $n = 3$ one could simply re-write those steps and replace 3 with n to obtain the expression. This is why i didn't include the full derivation as we are given the expression in the tutorial so I assume this isn't required but I include it for completeness.